

Peter Torpis

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EDUCATION

Jönköping International Business School

B.Sc. International Economics

Jönköping, Sweden

Aug 2023 — June 2026

- Relevant Coursework: Intermediate Micro/Macroeconomics, Statistical Methods, Econometrics I & II
- Extracurricular: Analyst, then Vice Head of Portfolio Management at the university's student finance club

PROJECTS

Market Microstructure Simulator (github.com/ptorpis/market-sim.git)

Jan 2026 — Present

- Built a discrete-time, event-driven market simulation framework in **C++23** with multiple agent archetypes (noise trader, market maker, informed trader); adapted the matching engine from MiniExchange with slight modifications
- Designed a strong type system via **CRTP** (**P**rice, **Q**uantity, **O**rdersID, etc.) for compile-time semantic correctness; tested across compilers (GCC & Clang) under **ASan**, **UBSan**, and **Valgrind**
- Developed a **Python** analysis suite featuring an interactive order book visualizer (plot, animate, step-through replay) and a simulation harness supporting **parameter sweeps**. Using the platform as a research tool to study market microstructure theory; conducted experiments replicating **Glosten-Milgrom (1985)** equilibrium spread predictions, with further extensions planned

MiniExchange: Financial Exchange Simulator (github.com/ptorpis/MiniExchange.git)

June 2025 — Jan 2026

- Engineered a low-latency limit order book exchange in **C++23**, achieving **<1μs median matching latency** and **230k+ requests/s** throughput with 1000 concurrent clients; implemented FIFO price-time priority matching with market/limit orders, modifications, cancellations, and self-trade prevention via template-based policies
- Designed a custom binary protocol with 8-byte aligned message frames over **TCP** and **UDP multicast** feed for real-time market data distribution to subscribers
- Profiled with **perf**, **Valgrind**, and **FlameGraph** to optimize cache locality and eliminate branch mispredictions; documented architecture and optimizations in a [technical paper series](#)

EXPERIENCE

Vice Head of Portfolio Management

JSA Investment Club

May 2025 — Present

Jönköping, Sweden

- Led research on portfolio allocation models (**Markowitz**, **Risk-Parity**, **Black-Litterman**); published "[Weaknesses in Statically Optimized Portfolios](#)"; demonstrated issues when naively applying models through empirical analysis. Conducting ongoing research to publish a follow-up to the report above
- Co-led & designed **Excel workshops**, as part of the university's Basic Financial Accounting and Human Resources courses, taught the basics of formulas, worksheets, and references to **450+ students**
- Overseeing **portfolio rebalancing** and performance reporting for the club's fund

Analyst

JSA Investment Club

September 2023 — May 2024

Jönköping, Sweden

- Developed foundational understanding of **valuation techniques** (DCF, Comparable Company Analysis) and financial modeling by conducting fundamental equity research and participating in internal equity pitch competitions

SKILLS

- **Programming Languages:** C++, Python
- **Technologies:** CMake + GCC, Linux, Git, GDB, GoogleTest, Pybind11, TCP/UDP, Claude Code, Wireshark
- **Languages:** English (Cambridge C1 + active use), Hungarian (native), Swedish (basic)